

LEINEFELDE, Germany

WMO: 104490

Lat: 51.3933N

Lon: 10.3122E

Elev: 353

StdP: 97.15

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.5	-10.7	-16.2	0.9	-12.4	-13.6	1.2	-9.9	12.4	4.8	11.2	3.3	2.6	90	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	27.3	18.3	25.4	17.5	23.8	16.7	19.3	25.1	18.3	23.7	17.4	22.4	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.8	21.5	16.2	12.1	20.4	15.4	11.4	19.5	56.4	25.2	53.0	23.7	50.2	22.5	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.8	7.6	-14.1	31.1	5.2	1.6	-17.9	32.3	-20.9	33.2	-23.9	34.0	-27.7	35.2		
(5)			-14.4	21.4	5.2	1.3	-18.1	22.4	-21.2	23.1	-24.1	23.8	-27.8	24.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-0.3	-0.7	3.5	6.7	11.4	14.2	16.6	16.7	13.2	8.9	3.8	1.3	(6)	
(7)		DBStd	7.31	5.25	5.21	4.13	3.79	3.65	3.49	3.32	3.25	3.11	3.65	3.86	3.92	(7)	
(8)		HDD10.0	1492	318	301	205	113	28	4	0	0	6	63	187	269	(8)	
(9)		HDD18.3	3827	576	534	461	350	214	132	75	72	157	292	436	527	(9)	
(10)		CDD10.0	762	0	1	2	13	72	129	206	208	101	29	1	0	0	(10)
(11)		CDD18.3	55	0	0	0	0	1	7	22	22	3	1	0	0	0	(11)
(12)		CDH23.3	459	0	0	0	0	8	65	185	182	18	1	0	0	0	(12)
(13)		CDH26.7	77	0	0	0	0	0	9	28	40	0	0	0	0	0	(13)
(14)	Wind	WSAvg	3.2	4.1	3.7	3.6	3.2	2.9	2.6	2.5	2.4	2.8	3.2	3.5	3.5	(14)	
(15)	Precipitation	PrecAvg	692	55	48	51	42	64	64	73	64	58	52	61	60	(15)	
(16)		PrecMax	940	115	99	133	75	144	104	120	133	129	159	111	132	(16)	
(17)		PrecMin	521	1	15	10	5	14	31	18	23	7	9	1	24	(17)	
(18)		PrecStd	108	28	25	30	15	33	19	28	27	31	32	27	26	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	12.8	16.9	20.1	24.3	28.1	29.2	30.6	25.4	21.1	14.1	11.5	(19)	
(20)			MCWB	8.1	9.6	10.2	12.1	15.7	18.6	19.3	19.0	16.7	15.0	10.6	9.0	(20)	
(21)		2%	DB	8.3	10.3	13.1	17.7	22.5	25.2	27.2	27.4	23.2	17.8	11.6	9.8	(21)	
(22)			MCWB	6.8	8.0	9.0	11.3	14.8	17.7	18.5	17.9	16.2	13.7	9.5	8.2	(22)	
(23)		5%	DB	7.2	7.6	10.6	15.5	20.5	23.1	25.5	25.2	20.7	15.8	10.2	8.2	(23)	
(24)			MCWB	5.9	5.8	7.6	10.0	13.9	16.4	17.8	17.3	15.4	12.5	8.8	7.0	(24)	
(25)		10%	DB	5.8	5.5	8.9	13.1	18.6	20.8	23.5	23.2	18.5	14.0	8.9	6.8	(25)	
(26)			MCWB	4.5	3.9	6.6	8.8	13.1	15.5	16.9	16.5	14.4	11.6	7.6	5.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	10.0	10.6	13.3	17.3	20.3	20.9	20.8	18.1	16.0	11.2	9.4	(27)	
(28)			MCDWB	9.4	12.6	15.2	18.3	22.2	26.0	27.5	27.0	23.4	19.5	12.8	10.6	(28)	
(29)		2%	WB	7.1	8.0	9.2	11.7	15.7	18.4	19.5	18.9	16.8	14.4	9.9	8.3	(29)	
(30)			MCDWB	8.3	9.8	12.6	16.5	21.1	23.6	25.5	25.0	21.3	17.0	11.2	9.6	(30)	
(31)		5%	WB	5.9	5.8	8.1	10.5	14.4	17.2	18.4	18.0	15.8	13.0	8.9	7.1	(31)	
(32)			MCDWB	7.0	7.4	10.3	14.5	19.5	21.9	23.8	23.7	19.5	15.0	10.3	8.3	(32)	
(33)		10%	WB	4.6	4.2	6.8	9.2	13.3	16.0	17.4	17.1	14.9	11.9	7.7	5.8	(33)	
(34)			MCDWB	5.8	5.4	8.6	12.7	17.8	20.0	22.3	21.9	18.3	13.7	8.9	6.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	5.3	6.3	8.0	9.1	8.6	9.2	9.4	7.8	6.4	4.8	3.8	(35)	
(36)			MCDBR	4.7	7.4	9.5	11.4	12.3	12.2	12.5	12.6	11.4	9.1	6.9	5.0	(36)	
(37)			MCWBR	4.3	5.7	6.4	6.5	6.6	6.2	6.0	5.6	5.8	5.5	5.3	4.5	(37)	
(38)		5% WB	MCDWB	4.8	7.1	8.7	10.3	11.3	11.0	11.6	11.2	9.8	8.0	6.4	4.7	(38)	
(39)			MCWBR	4.4	5.6	6.0	6.4	6.4	6.1	6.1	5.4	5.4	5.3	5.2	4.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.325	0.361	0.399	0.402	0.400	0.415	0.408	0.373	0.346	0.325	0.298	(40)		
(41)		taud	2.403	2.369	2.299	2.241	2.261	2.287	2.249	2.290	2.373	2.440	2.428	2.397	(41)		
(42)		Ebn at Noon	719	798	830	837	852	856	834	820	810	764	682	659	(42)		
(43)		Edh at Noon	60	81	105	124	128	126	129	118	97	75	59	52	(43)		
(44)	All-Sky Solar Radiation	RadAvg	0.72	1.42	2.50	4.00	4.87	5.27	4.96	4.25	3.07	1.78	0.82	0.55	(44)		
(45)		RadStd	0.10	0.20	0.31	0.49	0.56	0.45	0.66	0.41	0.40	0.20	0.11	0.07	(45)		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (40 neighbors)		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page